

REQUIREMENTS

What we Need

- Raspberry Pi 2 or 3
- Sd Card 8GB Class 10
- Arduino Uno
- HDMI splitter
- HDMI Selector
- HDMI2AV Converter (1080p supported)
- USB Grabber
- 5V 10A power supply
- HDMI Cables
- WS2812B RGB LED strip
- Wires

Software

- Raspberry Pi 2 / 3 Disk Image
- Arduino Sketch (<http://dgit.in/ArduinoSketch>)
- Hypercon.jar
- Arduino NeoPixel Library
- Win32 Disk imager

the wall behind your TV, matching the colours being displayed on the TV. Thus, if the majority of the screen was red, then the light emitted would be red, and so on.

Over the years this has been refined to multiple colors, matching small parts of the screen, and the aim is to make the TV watching experience even more immersive than it is already. It's a feature that most people like, but some might not, but it never fails to wow people the first time they see it. Tell them you built that yourself, and they're going to be even more impressed! Of course you could also just buy a Philips Ambilight TV, but what's geek about that?

CONFIGURING THE LEDS

Start From the bottom left corner, place the LEDs counterclockwise around the border of the TV and connect them at the corners (except the first one, obviously) with conductive wire – make sure that you connect the corresponding terminals on each LED correctly.

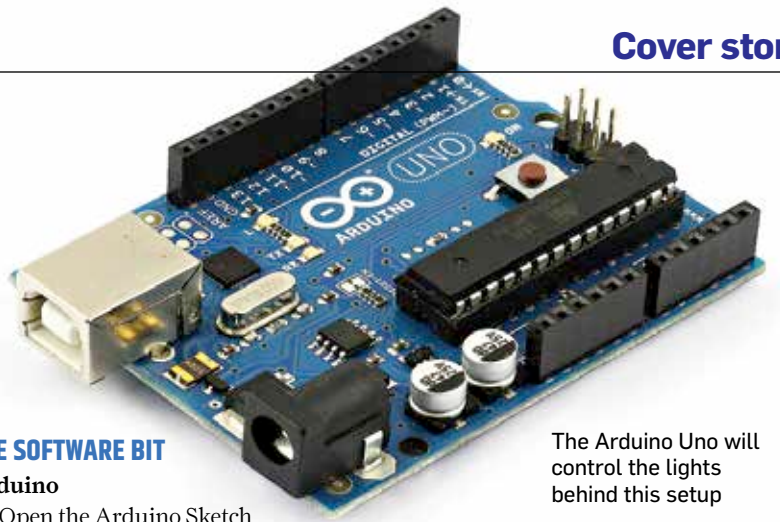
POWER SUPPLY

It is important to connect the GND from the Power supply to the Arduino

The connection to the Arduino is:

- LED Din to Arduino Pin (as configured in the Arduino Sketch)
- Power Supply GND to Arduino GND

Note: Count the number of Vertical LEDS and make a note of this.



THE SOFTWARE BIT

Arduino

1. Open the Arduino Sketch and change the number of LEDCOUNT to match the total number of LEDS we noted earlier.
2. Also in the line: `const char prefix [] = {0x41, 0x64, 0x61, 0x00, 0x57, 0x02};` we need to change the last two bytes. (0x57 and 0x02) to match your setup.
3. For this take the total number of LEDS on your system and subtract 1.
4. Convert this number to HEX (use Google calculator or any online decimal to hex converter) and replace the first byte (0x57) to 0x?? to match yours in the sketch.
5. Now go to XOR converter (xor.pw) and enter the following:
6. In the first Line enter 55
7. In the second line enter the byte you calculated in the above step (0x57 in our case)
8. Click Calculate XOR
9. Replace the last byte of your line by the Output you get. Make sure to include "0x" before the number (0x02) as this is represented in HEX format.
10. Also note down the Baud rate – 500000 in our sketch.
11. Now connect the D10 pin on the arduino to the Din of the LED – the third wire from the LED strip.
12. Download and include the Neopixel library (see box Requirements) by going to Tools > include Library > Add Zip file in the arduino software.
13. Save the sketch and compile for any errors, and then upload to Arduino.

INSTALLING THE IMAGE ON THE RASPBERRY PI SD CARD,

1. Unrar the OSMC_Pulse_setup_all_working.rar provided with the link.

The Arduino Uno will control the lights behind this setup

CONNECTING ALL THE EQUIPMENT

1. Plug the Arduino Pin 10 into the LED astrip data line and Arduino GND pin to V- of the power supply,
2. Plug the USB of the Arduino into port 1 of the Raspberry Pi.
3. Plug an HDMI cable from the HDMI splitter output 1 into your TV's HDMI-in
4. Plug the output of the HDMI Selector into the input of the HDMI splitter
5. Plug another HDMI cable from the HDMI splitter output2 into the HDMI2 AV converter.
6. Plug all the Devices into HDMI selector
7. Plug the video (yellow) cable to



Smartphone Master Race

Real phone pros can use their phones as well as normal people use desktops. From adblockers to VPN here's a list of tweaks no "smart" phone user should be without

<http://dgit.in/SPMstrRace>